

Unit7: Multimedia Design (4 Hrs.)

Development Phases and Development Teams; Analysis, Design, Development and Implementation Phase; Evaluation and Testing Phase

Learning Objectives

After studying this chapter, students will be able to:

- Define **Multimedia Design** and explain its importance.
- Describe the **Multimedia Development Life Cycle (MDLC)**.
- Explain the roles of **Multimedia Development Teams**.
- Understand each phase of multimedia development:
 - Analysis
 - Design
 - Development
 - Implementation
 - Evaluation and Testing
- Explain multimedia project management with practical examples.

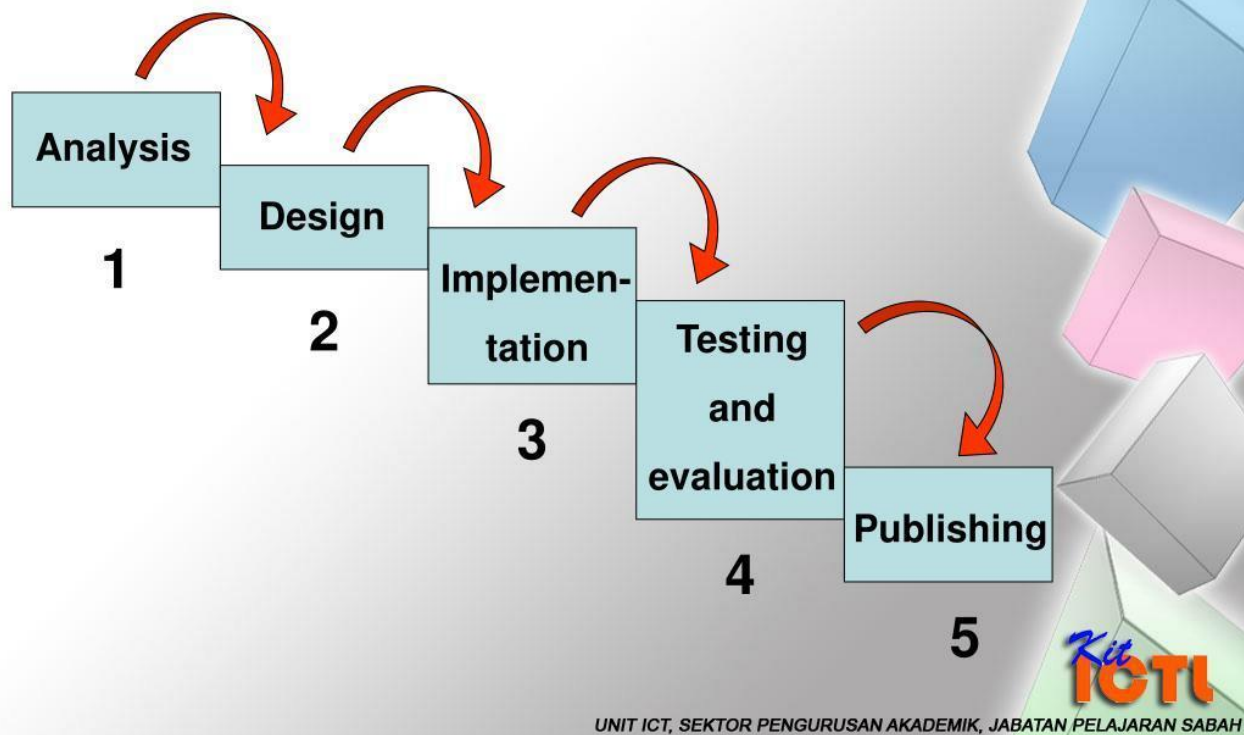
Summary

Topic	Key Points
Multimedia Design	Planning and integrating multimedia elements into an interactive application
Development Team	Project manager, analyst, designers, programmers, QA, database developer, client
Analysis Phase	Gather requirements and assess feasibility
Design Phase	Create UI, storyboard, navigation, and database design
Development Phase	Build and integrate multimedia content and application code
Implementation Phase	Deploy the system, train users, and migrate data
Evaluation & Testing Phase	Verify functionality, usability, performance, compatibility, and security

Chapter: Multimedia Design (4 Hours)



Steps in Multimedia Development

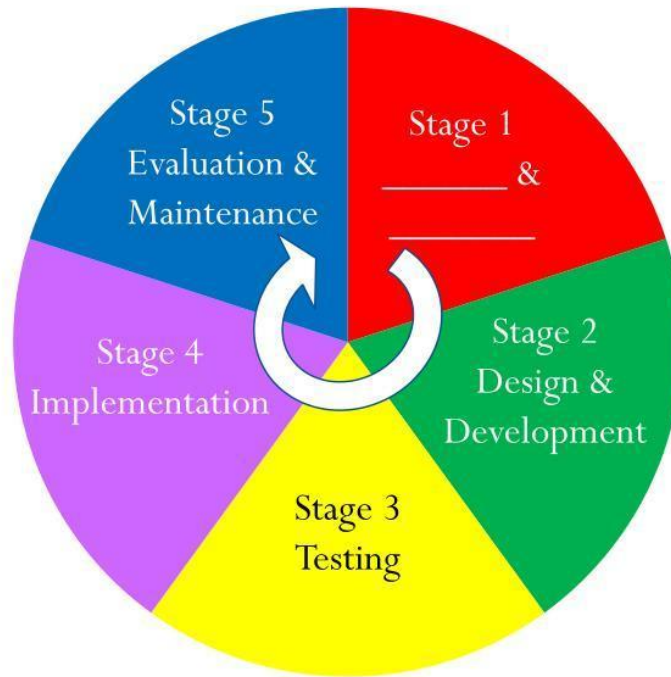


UNIT ICT, SEKTOR PENGURUSAN AKADEMIK, JABATAN PELAJARAN SABAH

Phase	Activities	Output
Analysis	Gather requirements from users	Software Requirements Specification
Design	Create UI, storyboard, and navigation	Design documents
Development	Develop videos, quizzes, animations, and code	Working application
Implementation	Install system and train users	Operational LMS
Evaluation & Testing	Test features, usability, performance, and security	Approved final system



Multimedia Project Life Cycle



5

1. Introduction to Multimedia Design

What is Multimedia Design?

Multimedia Design is the process of planning, creating, organizing, and integrating different multimedia elements such as **text, graphics, images, audio, video, animation, and interactive components** into a single application to communicate information effectively.

It focuses not only on the appearance of the multimedia application but also on its usability, functionality, and user experience.

Definition

Multimedia Design is the systematic process of analyzing user requirements, designing interfaces, creating multimedia content, developing applications, testing functionality, and deploying interactive multimedia systems.

Example

A university develops an **E-learning System**.

The system contains:

- Text notes
- Images
- Audio lectures
- Video tutorials
- Animations
- Interactive quizzes

All these multimedia components are carefully designed and integrated to provide an engaging learning experience.

Characteristics of Multimedia Design

- Interactive
 - User-friendly
 - Attractive interface
 - High usability
 - Efficient navigation
 - Responsive design
 - Cross-platform compatibility
-

Objectives of Multimedia Design

- Present information effectively.
- Improve learning and understanding.
- Increase user engagement.
- Simplify complex information.
- Enhance user satisfaction.

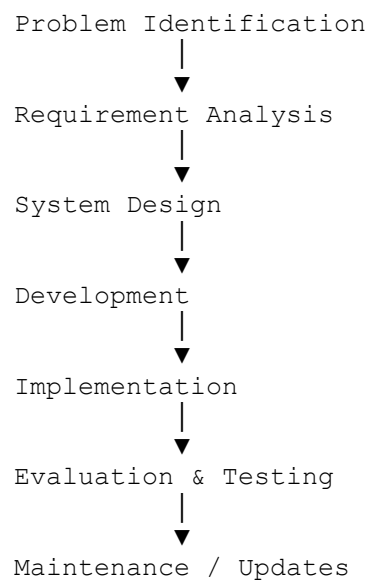
- Support interaction between users and the system.

Importance of Multimedia Design

Importance	Description
Better Communication	Combines multiple media types for effective communication
Interactive Learning	Improves student engagement
Better User Experience	Creates attractive and intuitive interfaces
Marketing	Produces effective advertisements and presentations
Entertainment	Used in games, movies, and animations
Business	Supports training, demonstrations, and product visualization

Multimedia Development Life Cycle (MDLC)

The Multimedia Development Life Cycle consists of several structured phases.



Case Study: Development of an E-Learning System

MDLC Stage	Activities	Example Output
Analysis	Gather requirements from students and teachers	Software Requirements Specification (SRS)
Design	Create storyboards, UI layouts, navigation, and database schema	Prototype and design documents
Development	Build the LMS, integrate multimedia, write code	Functional e-learning application
Implementation	Install the LMS, migrate data, and train users	Live LMS for students and faculty
Evaluation & Testing	Perform functional, usability, performance, compatibility, security, and UAT testing	Tested and approved system
Maintenance	Fix bugs, improve features, update security, and add enhancements	Updated versions with improved functionality

Stage 1: Analysis Phase

Definition

The **Analysis Phase** is the first stage of MDLC. In this phase, developers identify the problem, understand user requirements, define project objectives, and determine whether the project is feasible.

This phase answers the question:

"What should the multimedia application do?"

Objectives

- Understand user needs.
 - Define project scope.
 - Identify target audience.
 - Study existing systems.
 - Assess feasibility.
-

Activities

Requirement Collection

Information is gathered through:

- Interviews
 - Questionnaires
 - Meetings
 - Observation
 - Document analysis
-

Audience Analysis

Questions include:

- Who will use the system?
 - What is the users' age group?
 - What language should be used?
 - What devices will they use?
 - What accessibility features are needed?
-

Feasibility Study

The project is evaluated for:

- Technical feasibility
 - Economic feasibility
 - Operational feasibility
 - Schedule feasibility
 - Legal feasibility
-

Deliverables

- Software Requirement Specification (SRS)
 - Feasibility Report
 - Project Scope
 - Project Schedule
-

Example

Project: University E-Learning System

Requirements identified:

- Student login
- Video lectures
- Assignments
- Online quizzes
- Discussion forums
- Progress reports

These requirements are documented before moving to the next phase.

Stage 2: Design Phase

Definition

The **Design Phase** transforms the collected requirements into a detailed blueprint for the multimedia application.

It answers:

"How will the system work?"

Objectives

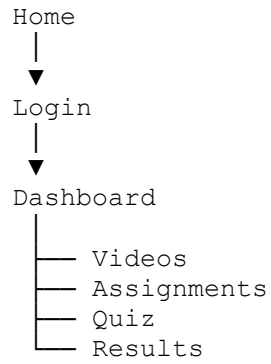
- Design the user interface.
 - Plan system architecture.
 - Design navigation.
 - Prepare database structure.
 - Plan multimedia elements.
-

Activities

Storyboard Design

The sequence of screens is sketched.

Example:



User Interface (UI) Design

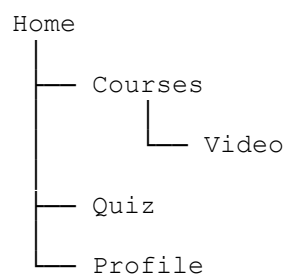
Design:

- Buttons
- Menus
- Icons
- Colors
- Fonts
- Layout

Navigation Design

Determine how users move through the application.

Example:



Database Design

Create tables such as:

- Students
- Teachers
- Courses

- Videos
- Quiz Results

Deliverables

- Storyboards
- Wireframes
- UI Mockups
- Navigation Structure
- Database Design

Example

A prototype of the e-learning portal is created, showing the login page, dashboard, course pages, and quiz interface before coding begins.

Stage 3: Development Phase

Definition

The **Development Phase** is where the actual multimedia application is built according to the design.

This answers:

"How can we build the system?"

Activities

Multimedia Content Creation

Develop:

- Images
- Videos
- Audio
- Animations
- Graphics
- Text

Programming

Develop application modules.

Programming languages may include:

- Java
- Python
- C#
- PHP
- JavaScript

Multimedia Integration

Integrate:

- Images
- Videos
- Audio
- Animation
- Database
- User Interface

Database Development

Create:

- Tables
- Relationships
- Queries
- Reports

Deliverables

- Working software
 - Source code
 - Multimedia assets
 - Database
-

Example

Developers create:

- Student registration page
- Video streaming module
- Quiz system
- Progress tracking dashboard

All components are integrated into one application.

Stage 4: Implementation Phase

Definition

The **Implementation Phase** is the stage where the completed multimedia application is deployed and made available to users.

It answers:

"How will users access the system?"

Activities

Software Installation

Install the application on:

- Computers
 - Mobile devices
 - Servers
-

User Training

Provide:

- User manuals
- Video tutorials
- Workshops
- Help documents

Data Migration

Transfer existing data into the new system.

Example:

Import student records from the old database into the new LMS.

Deployment

Launch the system for actual users.

Deliverables

- Installed application
 - User manuals
 - Training materials
 - Operational system
-

Example

The university deploys the LMS on its server. Students receive usernames and passwords, and faculty members attend training sessions before the semester begins.

Stage 5: Evaluation and Testing Phase

Definition

The **Evaluation and Testing Phase** ensures that the multimedia application is functioning correctly, satisfies user requirements, and is ready for use.

It answers:

"Does the application work correctly?"

Objectives

- Identify defects.
 - Improve quality.
 - Verify requirements.
 - Ensure user satisfaction.
-

Types of Testing

Functional Testing

Verifies all features work correctly.

Example:

- Login
 - Video playback
 - Quiz submission
 - Report generation
-

Usability Testing

Checks whether the system is easy to use.

Questions include:

- Are menus easy to understand?
 - Is navigation simple?
 - Are buttons clearly visible?
-

Performance Testing

Measures:

- Response time
 - Loading speed
 - Server performance
 - Scalability
-

Compatibility Testing

Checks compatibility with:

- Windows
- Linux
- Android
- iOS
- Chrome
- Firefox
- Edge

Security Testing

Tests:

- User authentication
- Password encryption
- Access control
- Protection against unauthorized access

User Acceptance Testing (UAT)

End users test the application before official release.

Deliverables

- Test Plan
- Test Cases
- Bug Reports
- Test Results
- User Feedback
- Final Approval

Example

Teachers and students test the LMS. They report that quizzes work correctly, but some videos load slowly. Developers optimize video compression and server settings before the final release.

Stage 6: Maintenance Phase

Definition

The **Maintenance Phase** begins after the system has been deployed. Its purpose is to keep the application functional, secure, and up to date.

It answers:

"How do we keep the system useful over time?"

Activities

Corrective Maintenance

Fix software bugs and errors.

Example:

Resolve a login issue reported by users.

Adaptive Maintenance

Modify the system to work with new technologies.

Example:

Update the application to support the latest Android version.

Perfective Maintenance

Improve existing features.

Example:

Add dark mode or improve dashboard performance.

Preventive Maintenance

Reduce the risk of future problems.

Example:

Update security patches and optimize the database.

Deliverables

- Updated software
 - Bug fixes
 - Performance improvements
 - Security updates
 - New features
-

Example

After one year, the university adds:

- Live virtual classrooms
- AI-based course recommendations
- Mobile notifications
- Faster search functionality

These enhancements are part of the maintenance phase.

2. Multimedia Development Phases and Development Teams

A multimedia project requires specialists from different fields working together.

Typical Multimedia Development Team

Multimedia Team Members

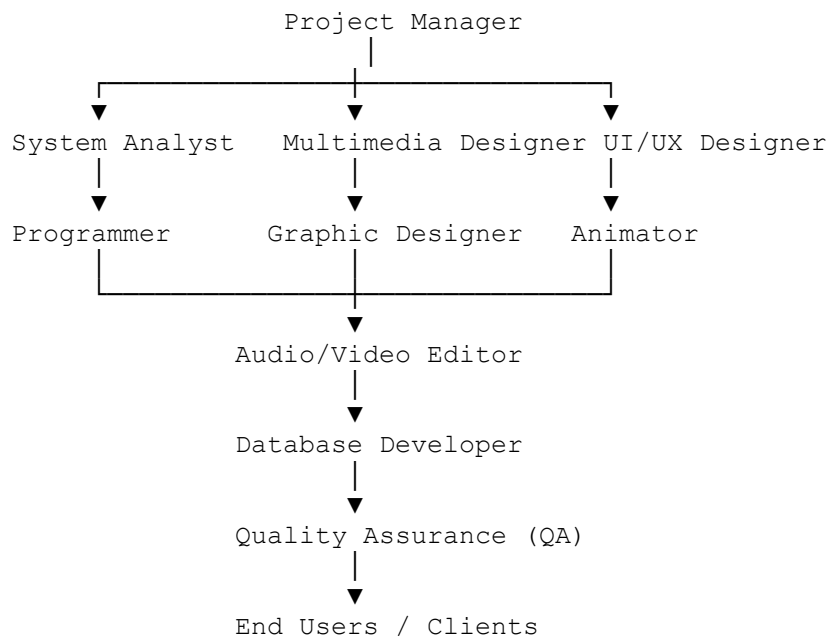
A multimedia team member consists of the following:

- Project manager
- Multimedia designer
- Interface designer
- Multimedia programmer
- Computer programmers
- Writer • Subject matter expert
- Audio specialist
- Video specialist
- Producer for the web
- Permission specialist





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Roles of Development Team Members

1. Project Manager

Responsibilities

- Plans the project

- Allocates resources
- Manages budget
- Schedules activities
- Monitors progress
- Communicates with clients

Skills

- Leadership
 - Communication
 - Risk management
 - Decision making
-

2. System Analyst

Responsibilities

- Collects user requirements
 - Studies existing systems
 - Prepares feasibility reports
 - Creates system specifications
-

3. Multimedia Designer

Responsible for:

- Overall visual appearance
 - Layout design
 - Multimedia integration
 - Navigation structure
-

4. Graphic Designer

Creates:

- Logos
- Icons
- Images
- Illustrations
- User interface graphics

Tools:

- Adobe Photoshop
 - Adobe Illustrator
 - CorelDRAW
 - Canva
-

5. Animator

Creates:

- 2D animation
- 3D animation
- Motion graphics
- Character animation

Software:

- Adobe Animate
 - Blender
 - Maya
-

6. Audio and Video Specialist

Produces:

- Narration
- Music
- Video editing
- Sound effects

Tools:

- Audacity
 - Adobe Audition
 - Premiere Pro
 - DaVinci Resolve
-

7. Programmer

Develops:

- Multimedia software
- Interactive applications
- Database connectivity

- User interaction

Languages:

- Java
 - C#
 - Python
 - JavaScript
 - PHP
-

8. Database Developer

Manages:

- User information
 - Multimedia content
 - Application data
 - Security
-

9. Quality Assurance (QA) Engineer

Tests:

- Performance
 - Functionality
 - Compatibility
 - Security
 - Usability
-

10. Client / End User

Provides:

- Requirements
 - Feedback
 - Acceptance testing
-

3. Analysis Phase

Definition

The **Analysis Phase** is the first and one of the most important stages of multimedia development. During this phase, developers study the problem, understand user requirements, identify project goals, analyze available resources, and determine whether the project is technically and economically feasible.

Objectives

- Understand user needs.
 - Identify target audience.
 - Analyze existing systems.
 - Define project scope.
 - Conduct feasibility studies.
-

Activities

Requirement Gathering

Methods:

- Interviews
 - Questionnaires
 - Observation
 - Document analysis
-

Audience Analysis

Questions include:

- Who will use the application?
 - What is their age group?
 - What language do they prefer?
 - What devices will they use?
-

Feasibility Study

1. Technical feasibility
2. Economic feasibility
3. Operational feasibility
4. Schedule feasibility

5. Legal feasibility

Deliverables

- Requirement Specification (SRS)
 - Feasibility Report
 - Project Plan
-

Example

A school wants an e-learning application.

Requirements include:

- Video lectures
- Online quizzes
- Student login
- Progress reports

These requirements are documented during the analysis phase.

4. Design Phase

Definition

The **Design Phase** converts the requirements collected during analysis into a detailed blueprint for the multimedia application.

Objectives

- Plan the application structure.
 - Design interfaces.
 - Organize multimedia elements.
 - Define navigation.
 - Prepare database design.
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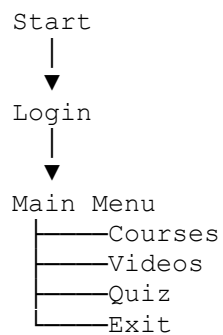
Activities

Storyboarding

Creates a visual sequence of screens.

Flowchart Design

Shows application logic.



User Interface (UI) Design

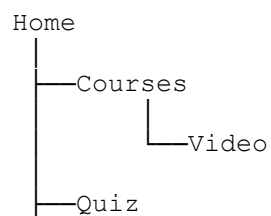
Designs:

- Buttons
 - Menus
 - Icons
 - Layout
 - Colors
 - Typography
-

Navigation Design

Defines movement between screens.

Example:



Database Design

Defines:

- Tables
 - Relationships
 - Data storage
-

Deliverables

- Storyboards
 - Wireframes
 - UI mockups
 - Navigation diagrams
 - Database schema
-

5. Development Phase

Definition

The **Development Phase** is where the multimedia application is actually built according to the design.

Activities

Content Creation

Develop:

- Images
 - Audio
 - Video
 - Animation
 - Text
-

Programming

Write application code.

Example languages:

- Java
- Python
- JavaScript
- C#

Multimedia Integration

Combine:

- Images
- Videos
- Audio
- Database
- Animation

Database Development

Create:

- Tables
- Queries
- Relationships

Deliverables

- Working multimedia application
- Source code
- Multimedia assets
- Database

Example

An e-learning application now includes:

- Login page
 - Video player
 - Quiz module
 - Student dashboard
 - Database connectivity
-

6. Implementation Phase

Definition

The **Implementation Phase** is the process of deploying the completed multimedia application for real users.

Activities

Installation

Deploy software on:

- Computers
 - Mobile devices
 - Web servers
-

User Training

Provide:

- User manuals
 - Training sessions
 - Video tutorials
-

Data Migration

Transfer data from old systems to the new application.

System Deployment

Launch the application.

Example:

A university launches its Learning Management System (LMS) for students and teachers.

Deliverables

- Installed system
 - User documentation
 - Training materials
-

7. Evaluation and Testing Phase

Definition

The **Evaluation and Testing Phase** ensures that the multimedia application functions correctly, meets user requirements, and provides a high-quality user experience before and after deployment.

Objectives

- Detect errors.
 - Improve quality.
 - Verify requirements.
 - Ensure user satisfaction.
-

Types of Testing

1. Functional Testing

Checks whether every feature works correctly.

Examples:

- Login
 - Registration
 - Quiz submission
-

2. Usability Testing

Evaluates:

- Ease of use
 - Navigation
 - User satisfaction
-

3. Performance Testing

Measures:

- Speed
 - Response time
 - Resource usage
-

4. Compatibility Testing

Checks operation on different:

- Browsers
 - Operating systems
 - Screen sizes
 - Mobile devices
-

5. Security Testing

Tests:

- Authentication
 - Authorization
 - Data protection
-

6. User Acceptance Testing (UAT)

End users verify that the system meets their requirements before final acceptance.

Evaluation Criteria

Criterion	Purpose
Functionality	Verify required features
Usability	Ensure ease of use
Reliability	Check stable operation
Efficiency	Measure performance
Maintainability	Ease of updates
Portability	Ability to run on different platforms

Deliverables

- Test plan
 - Test cases
 - Bug reports
 - Test results
 - User feedback
 - Final approval
-

Practical Example: Multimedia E-Learning System

Phase	Activities	Output
Analysis	Gather requirements from teachers and students	Software Requirements Specification
Design	Create UI, storyboard, and navigation	Design documents
Development	Develop videos, quizzes, animations, and code	Working application
Implementation	Install system and train users	Operational LMS
Evaluation & Testing	Test features, usability, performance, and security	Approved final system

Advantages of Following the Multimedia Development Process

- Reduces development risks.
- Improves software quality.
- Enhances teamwork and communication.
- Detects errors early.
- Improves user satisfaction.
- Simplifies future maintenance and upgrades.